

```
1  /*
2  यदा यदा हि धर्मस्य क्लानिर्भवति भारत।
3  अभ्युत्थानमुन्मार्शस्य तदात्मानं सृजाम्यहम्॥
4  */
5  /*
6  ॐ त्र्यम्बकं यजामहे सुगन्धिं पुष्टिवर्धनम् ।
7  उर्वारुकमिव बभूवन्मृत्योर्मुक्षीय मामृतात् ।
8  */
9
10 #include <bits/stdc++.h>
11 using namespace std;
12
13 bool isposs(vector<int> &v1, int tar, int k) {
14     int sub_sum = 0, sum = 0, n = v1.size();
15     for (int i = 0; i < n; i++) {
16         if (sum + v1[i] <= tar) {
17             sum += v1[i];
18         } else {
19             sub_sum++;
20             sum = v1[i];
21         }
22     }
23     return sub_sum <= k;
```

```
Testcase #1
1 5 1
2 7 2 5 10 8

1 18

Testcase #2
1 7 2
2 2 3 1 2 4 3 5

1 8

Run Task_1
Test Results
2 tests passed 2 tests total, 68ms

Testing started at 18:57 ...
Building "Task_1" ...
Build completed in 1593ms

-----[ Testcase #1 ]-----
18
```

```
13 bool isposs(vector<int> &v1, int tar, int k) {
23     return sub_sum <= k;
24 }
25
26 int32_t main() {
27     ios_base::sync_with_stdio(false);
28     cin.tie(&tiestr);
29     cout.tie(&tiestr);
30     cin.exceptions(exception::istream::failbit);
31     int n, k, max_n = INT_MIN, sum = 0;
32     cin >> n >> k;
33     vector<int> v1(n);
34     for (int i = 0; i < n; i++) {
35         cin >> v1[i];
36         sum += v1[i];
37         max_n = max(max_n, v1[i]);
38     }
39     int lo = max_n - 1, hi = sum;
40     while (hi - lo > 1) {
41         int mid = lo + ((hi - lo) >> 1);
42         if (isposs(v1, mid, k)) {
43             hi = mid;
44         } else {
45             lo = mid;
```

```
Testcase #1
1 5 1
2 7 2 5 10 8

1 18

Testcase #2
1 7 2
2 2 3 1 2 4 3 5

1 8

Run Task_1
Test Results
2 tests passed 2 tests total, 68ms

-----[ Testcase #1 ]-----
18

[+] SUCCESS: CORRECT ANSWER
-----[ Testcase #2 ]-----
8
```

The image shows a C++ IDE with a code editor on the left and a test runner on the right. The code implements a binary search function to find the maximum element in an array.

```
int32_t main() {
    for (int i = 0; i < n; i++) {
        sum += v1[i];
        max_n = max(a, max_n, b, v1[i]);
    }
    int lo = max_n - 1, hi = sum;
    while (hi - lo > 1) {
        int mid = lo + ((hi - lo) >> 1);
        if (isposs(&v1, tar, mid, k)) {
            hi = mid;
        } else {
            lo = mid;
        }
    }
    cout << hi << endl;
    return 0;
}
```

The test runner on the right shows two test cases:

- Testcase #1: Input array [5, 1, 7, 2, 5, 10, 8], Target 18. Output: 18.
- Testcase #2: Input array [7, 2, 2, 3, 1, 2, 4, 3, 5], Target 8. Output: 8.

Both test cases passed successfully.